Interaction sample 1: Gradient updates

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_1(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_1(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_1(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

This document intentionally mixes prose, headings, short lists, and mathematics. Its PDF uses explicit page breaks to provide exactly twenty-five numbered pages. The continuous LyX editing view does not display PDF page boundaries, so compare the numbered section headings when repeating an interaction or reporting a delayed response.

1. Checkpoint 1. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 2: Step sizes

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_2(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_2(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_2(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 2. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 3: Residuals

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_3(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_3(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_3(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 3. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 4: Feasible points

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_4(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_4(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_4(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

This document intentionally mixes prose, headings, short lists, and mathematics. Its PDF uses explicit page breaks to provide exactly twenty-five numbered pages. The continuous LyX editing view does not display PDF page boundaries, so compare the numbered section headings when repeating an interaction or reporting a delayed response.

1. Checkpoint 4. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 5: Projections

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_5(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_5(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_5(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 5. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 6: Quadratic models

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_6(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_6(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_6(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 6. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 7: Coordinate updates

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_7(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_7(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_7(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 7. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 8: Stopping rules

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_8(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_8(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_8(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 8. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 9: Line searches

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_9(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_9(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_9(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 9. Select one sentence, then reverse the selection direction.
2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 10: Local models

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{10}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{10}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{10}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 10. Select one sentence, then reverse the selection direction.
2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 11: Norms and distances

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{11}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{11}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{11}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 11. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 12: Averaging

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{12}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{12}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{12}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 \,.$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 12. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 13: Perturbations

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{13}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{13}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{13}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 13. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 14: Conditioning

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{14}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{14}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{14}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 14. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 15: Constraints

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{15}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{15}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{15}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

This document intentionally mixes prose, headings, short lists, and mathematics. Its PDF uses explicit page breaks to provide exactly twenty-five numbered pages. The continuous LyX editing view does not display PDF page boundaries, so compare the numbered section headings when repeating an interaction or reporting a delayed response.

1. Checkpoint 15. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 16: Sparse vectors

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{16}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{16}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{16}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

This document intentionally mixes prose, headings, short lists, and mathematics. Its PDF uses explicit page breaks to provide exactly twenty-five numbered pages. The continuous LyX editing view does not display PDF page boundaries, so compare the numbered section headings when repeating an interaction or reporting a delayed response.

1. Checkpoint 16. Select one sentence, then reverse the selection direction.
2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 17: Dual variables

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{17}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{17}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{17}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 \,.$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

This document intentionally mixes prose, headings, short lists, and mathematics. Its PDF uses explicit page breaks to provide exactly twenty-five numbered pages. The continuous LyX editing view does not display PDF page boundaries, so compare the numbered section headings when repeating an interaction or reporting a delayed response.

1. Checkpoint 17. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 18: Error bounds

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{18}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{18}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{18}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 18. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 19: Descent estimates

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{19}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{19}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{19}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

This document intentionally mixes prose, headings, short lists, and mathematics. Its PDF uses explicit page breaks to provide exactly twenty-five numbered pages. The continuous LyX editing view does not display PDF page boundaries, so compare the numbered section headings when repeating an interaction or reporting a delayed response.

1. Checkpoint 19. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 20: Initialization

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{20}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{20}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{20}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

This document intentionally mixes prose, headings, short lists, and mathematics. Its PDF uses explicit page breaks to provide exactly twenty-five numbered pages. The continuous LyX editing view does not display PDF page boundaries, so compare the numbered section headings when repeating an interaction or reporting a delayed response.

1. Checkpoint 20. Select one sentence, then reverse the selection direction.
2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 21: Restart rules

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{21}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{21}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{21}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

This document intentionally mixes prose, headings, short lists, and mathematics. Its PDF uses explicit page breaks to provide exactly twenty-five numbered pages. The continuous LyX editing view does not display PDF page boundaries, so compare the numbered section headings when repeating an interaction or reporting a delayed response.

1. Checkpoint 21. Select one sentence, then reverse the selection direction.
2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 22: Parameter choices

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{22}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{22}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{22}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

This document intentionally mixes prose, headings, short lists, and mathematics. Its PDF uses explicit page breaks to provide exactly twenty-five numbered pages. The continuous LyX editing view does not display PDF page boundaries, so compare the numbered section headings when repeating an interaction or reporting a delayed response.

1. Checkpoint 22. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 23: Model comparisons

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{23}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{23}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{23}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 23. Select one sentence, then reverse the selection direction.
2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 24: Consistency checks

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{24}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{24}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{24}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

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1. Checkpoint 24. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.

Interaction sample 25: Final verification

Each section is a self-contained synthetic example for testing the editor. Read the paragraph, move the pointer across several lines, and select a phrase in either direction. Every sentence was written for this demonstration; no personal manuscript, unpublished result, or private file is included.

The model uses a vector of real variables and a smooth quadratic objective. The displayed expressions are native editable formula insets. Clicking inside a formula should place the mathematical cursor at the requested symbol. Scrolling past the equation should preserve its spacing, subscripts, and surrounding text.

$$f_{25}(x) = \frac{1}{2}\sum_{i=1}^{n}(x_i - b_i)^2, \qquad \nabla f_{25}(x) = x - b.$$

To test sustained scrolling, move through several sections without pausing, then reverse direction. Observe whether the content continues moving after the wheel has stopped. A delayed response can indicate queued input, while an increasingly irregular display may indicate work in frame capture, transport, or presentation.

To test a growing selection, begin near the top of the document and drag toward the lower edge of the viewport. Keep the button held as the editor scrolls into later sections. The highlighted range should follow the pointer, and releasing the button should end the gesture promptly.

$$x_{k+1} = x_k - \alpha_k \nabla f_{25}(x_k), \qquad r_k = \|x_{k+1} - x_k\|_2 .$$

Open the Review panel and repeat the same gesture. Reading annotations should not interrupt the active selection. After releasing the pointer, the review list and prepared clipboard may catch up. An explicit request to create a comment must still capture the complete selected passage at that moment.

This document intentionally mixes prose, headings, short lists, and mathematics. Its PDF uses explicit page breaks to provide exactly twenty-five numbered pages. The continuous LyX editing view does not display PDF page boundaries, so compare the numbered section headings when repeating an interaction or reporting a delayed response.

1. Checkpoint 25. Select one sentence, then reverse the selection direction.

2. Release the pointer, pause, and verify that scrolling and highlighting stop.